SEMINAR 3

Dezvoltări în serie Taylor

Fie $f : I \to \mathbb{R}$, $a \in I$ a.î. f indefinit derivabilă.

• Sn POLINOM TAYLOR de ordin n asociat fct. f în pct. a, polinomul:

$$T_n(x,a) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!} \cdot (x-a)^k =$$

$$= f(a) + \frac{f'(a)}{1!}(x-a) + \ldots + \frac{f^{(n)}(x)}{n!} \cdot (x-a)^n$$

• Sn. REST TAYLOR de ordin n asociat fct. f în pct. a:

$$R_n(\cdot,a) : I \to \mathbb{R},\ R_n(x,a) = f(x) - T_n(x,a)$$

Formula lui Taylor

$$f(x) = T_n(x,a) + R_n(x,a)$$

Sub forma lui Lagrange:

$$R_n(x,a) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1},$$ unde ξ se află în fct. de a și x.

• Sn. serie Taylor asoc. fct. f în pct. a, seria:

$$\sum_{n \geq 0} \frac{f^{(n)}(a)}{n!} \cdot (x-a)^n$$

$$a = 0 : \sum_{n \geq 0} \frac{f^{n}(0)}{n!} \cdot x^n$$

• Fie A_x mult. de conv. a acestei serii. Atunci:

$$f(x) = \sum_{n \geq 0} \frac{f^{n}(a)}{n!} \cdot (x-a)^n, \ \forall x \in I \cap A_x$$

cu prop. că $\lim_{n \to \infty} R_n(x,a) = 0$

Știu $\sum_{n\geq 0} \ldots = \frac{\ldots}{1-x}, x\in(-1,1)$

$$\sum_{n\geq 1} x^n = \frac{x}{1-x}, x\in(-1,1)$$

1. a) Dem. că $e^x = \sum_{n\geq 0} \frac{x^n}{n!}, \forall x\in\mathbb{R}$

b) Calc. val. nr. $\frac{1}{\sqrt{e}}$ cu 3 zecimale exacte.

c) Dezvoltați în serie de puteri $g(x) = e^{x^3}$

d) Calculați sumele seriilor

$$\sum_{n\geq 0} \frac{1}{n!}, \sum_{n\geq 0} \frac{an^2+bn+c}{n!}$$

DEM:

a) Fie $f:\mathbb{R}\to\mathbb{R}, f(x)=e^x$, f indefinit derivabilă $\forall x\in\mathbb{R}$.

- Seria de puteri asoc. fct f este:

$$\sum_{n\geq 0} \frac{f^{(n)}(0)}{n!}\cdot x^n$$

$f^{(n)}(x) = e^x, \forall n\in\mathbb{N}$

$\Rightarrow f^{(n)}(0)=1$

$\Rightarrow$ Seria de puteri asoc. fct. f este:

$$\sum_{n\geq 0} \frac{1}{n!}\cdot x^n.$$

$A_x=?$

$l = \overline{\lim} \left|\frac{a_{n+1}}{a_n}\right| = \lim \frac{1}{n+1} = 0 \Rightarrow R=\infty.$

$\Rightarrow A_x = (-\infty,\infty) = \mathbb{R}$

- $x=?$ $x\in\mathbb{R}$ a.î. $\lim_{n\to\infty} R_n(x,0)=0$

Form. Restului sub forma Lagrange

$R_n(x,0) = \frac{f^{(n+1)}(\xi)}{(n+1)!}\cdot x^{n+1}$, cu ξ între 0 și x.

$$|R_n(x,0)| = \left| \frac{e^{\xi}}{(n+1)!} x^{n+1} \right| =$$

$$= \frac{e^{\xi}}{(n+1)!} \cdot |x|^{n+1}$$

$$\frac{b_{n+1}}{b_n} = \frac{e^{\xi}}{(n+2)!} \cdot |x|^{n+1} \cdot \frac{(n+1)!}{e^{\xi}} \cdot \frac{1}{|x|^{n+1}}$$

$$= \frac{|x|}{n+2} \xrightarrow{n} 0 < 1 \Rightarrow b_n \to 0 \Rightarrow$$

$$\Rightarrow \lim_{n \to \infty} R_n(x,0) = 0$$

$$\Rightarrow \boxed{e^x = \sum_{n \geq 0} \frac{x^n}{n!}, \ \forall x \in \mathbb{R}.}$$

b) $\frac{1}{\sqrt{e}} \simeq \ldots, \_\_\_ \ldots$

$$e^{-\frac{1}{2}} = \sum_{n \geq 0} \frac{\left(-\frac{1}{2}\right)^n}{n!} =$$

$$= 1 + \frac{\left(-\frac{1}{2}\right)^1}{1!} + \frac{\left(-\frac{1}{2}\right)^2}{2!} + \ldots + \frac{\left(-\frac{1}{2}\right)^{100}}{100!} + \ldots$$

$n = ?$

$n \in \mathbb{N}$ cel mai mic posibil a.î.

$|R_n(x,0)| < 0{,}001.$

$x = -\frac{1}{2}$

Deci $|R_n(-\frac{1}{2},0)| = \left| \frac{e^{\xi}}{(n+1)!} \cdot \left(-\frac{1}{2}\right)^{n+1} \right| =$

$$= \frac{\overbrace{e^{\xi}}^{<1}}{(n+1)!} \cdot \left(\frac{1}{2}\right)^{n+1}, \ \xi \in \left(-\frac{1}{2}, 0\right)$$

$$< 0{,}001 = \frac{1}{1000}$$

$n = ?$

$$\frac{1}{(n+1)! \cdot 2^{n+1}} < \frac{1}{1000}$$

$$1000 < (n+1)! \cdot 2^{n+1}$$

$$500 < (n+1)! \cdot 2^n$$

$\Rightarrow \cancel{\boxed{n=3}}$ $\boxed{n=4}$

$$e^{-\frac{1}{2}} \approx 1 + \frac{\left(-\frac{1}{2}\right)}{1!} + \frac{\left(-\frac{1}{2}\right)^2}{2!} + \frac{\left(-\frac{1}{2}\right)^3}{3!} + \frac{\left(-\frac{1}{2}\right)^4}{4!}$$

$$= 1 - \frac{1}{2} + \frac{1}{8} - \frac{1}{24} + \frac{1}{16\cdot 24} = \underline{\underline{0,606}}$$

c) în (1) înlocuim $x \to x^3$.

$$e^{x^3} = \sum_{n\geq 0} \frac{(x^3)^n}{n!} = \sum_{n\geq 0} \frac{x^{3n}}{n!}, \forall x \in \mathbb{R}$$

d) $\sum_{n\geq 0} \frac{1}{n!}$

$$e^x = \sum_{n\geq 0} \frac{x^n}{n!}, \forall x \in \mathbb{R}$$

$$x = 1 \Rightarrow \sum_{n\geq 0} \frac{1}{n!} = e^1 = e.$$

(*) $\sum_{n\geq 0} \frac{an^2+bn+c}{n!} =$

$$= \sum_{n\geq 0}\left(\frac{an^2}{n!} + \frac{bn}{n!} + \frac{c}{n!}\right)$$

$$\sum_{n\geq 0} \frac{n}{n!} = 0 + \sum_{n\geq 1} \frac{n}{n!} = 0 + \sum_{n\geq 1} \frac{1}{(n-1)!}$$

$$= 0 + \sum_{n\geq 0} \frac{1}{n!} = e.$$

$$\sum_{n\geq 0} \frac{n^2}{n!} = 0 + \sum_{n\geq 1} \frac{n^2}{n!} = 0 + \sum_{n\geq 1} \frac{n}{(n-1)!}$$

$$= \sum_{n\geq 0} \frac{n+1}{n!} = \sum_{n\geq 0} \frac{n}{n!} + \sum_{n\geq 0} \frac{1}{n!} =$$

$$= e + e = 2e. \Rightarrow$$

$$\Rightarrow (*) = a\cdot 2e + b\cdot e + c\cdot e = e(2a+b+c).$$

2. Să se dezvolte în serie de puteri fct.

a) $f:\mathbb{R}\to\mathbb{R}, f(x) = \cos x$

b) $f:\mathbb{R}\to\mathbb{R}, f(x) = \sin x$

c) $f:\mathbb{R}\to\mathbb{R}, f(x) = \sin x^2$

d) $f:\mathbb{R}\to\mathbb{R}, f(x) = \cos^2 x$

SOL:

$$(*)\quad \cos x = \sum_{n\geq 0} \frac{(-1)^n \cdot x^{2n}}{(2n)!}, \forall x \in \mathbb{R}$$

$$(**)\quad \sin x = \sum_{n\geq 0} \frac{(-1)^n \cdot x^{2n+1}}{(2n+1)!}, \forall x \in \mathbb{R}$$

c) $$\sin x^2 = \sum_{n\geq 0} \frac{(-1)^n \cdot x^{4n+2}}{(2n+1)!}, \forall x \in \mathbb{R}$$

d) $$\cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 \Rightarrow$$

$$\Rightarrow \cos^2 x = \frac{\cos 2x + 1}{2} =$$

$$= \frac{1}{2} + \frac{1}{2} \cdot \cos 2x =$$

$$= \frac{1}{2} + \frac{1}{2} \cdot \sum_{n\geq 0} \frac{(-1)^n \cdot (2x)^{2n}}{(2n)!} =$$

$$= \frac{1}{2} + \sum_{n\geq 0} \frac{(-1)^n \cdot 2^{2n-1} \cdot x^{2n}}{(2n)!}$$

$$\boxed{\begin{array}{l} \dfrac{1}{1-x} = \sum_{n\geq 0} x^n, \forall x \in (-1,1) \\ \dfrac{x}{1-x} = \sum_{n\geq 1} x^n \\ \sin x = \ldots \\ \cos x = \ldots \\ e^x = \ldots \\ (1+x)^\alpha = 1 + \sum_{n\geq 1} \dfrac{\alpha(\alpha-1)(\alpha-2)\ldots(\alpha-n+1)}{n!} \cdot x^n, \forall x \in (-1,1) \\ \hookrightarrow \text{serie binomială} \end{array}}$$

3. Dezvoltați în serie de puteri ale lui x și precizați intervalul pe care dezv. este valabilă, f.d.:

$$f: \mathbb{R}\setminus\{1,3\} \to \mathbb{R}, \quad f(x) = \frac{3x-5}{x^2-4x+3}.$$

Obs $\sum_n a_n x^n + \sum_n b_n x^n = \sum_n (a_n + b_n) x^n$

$\sum_n a_n x^n \hookrightarrow R'$, $\sum_n b_n x^n \hookrightarrow R''$, $\sum_n (a_n+b_n)x^n \hookrightarrow \min\{R', R''\}$

SOL:

$$\frac{3x-5}{x^2-4x+3} = \frac{3x-5}{(x-1)(x-3)} =$$

$$= \frac{A}{x-1} + \frac{B}{x-3}$$

$3x - 5 = A(x-3) + B(x-1)$

$x = 3 : \ 4 = 2B \Rightarrow B = 2.$

$x = 1 : \ -2 = -2A \Rightarrow A = 1$

$$\Rightarrow f(x) = \frac{1}{x-1} + \frac{2}{x-3}.$$

$$\frac{1}{x-1} = -\frac{1}{1-x} = -\sum_{n\geq 0} x^n, \ \forall x \in (-1,1)$$

$x \to \frac{x}{3}$

$$\frac{2}{x-3} = -2 \cdot \frac{1}{3-x} = -\frac{2}{3} \cdot \frac{1}{1-\frac{x}{3}} =$$

$$= -\frac{2}{3} \cdot \sum_{n\geq 0} \left(\frac{x}{3}\right)^n, \ \forall \frac{x}{3} \in (-1,1) \Leftrightarrow$$

$\Leftrightarrow x \in (-3,3)$

$$\Rightarrow f(x) = -\sum_{n\geq 0} x^n - \frac{2}{3} \cdot \sum_{n\geq 0} \frac{x^n}{3^n} =$$

$$= \sum_{n\geq 0} \left(-1 - \frac{2}{3^{n+1}}\right) \cdot x^n,$$

$\forall x \in (-1,1)$

4. (a) Să se dezvolte în serie de puteri fd. $f : (-1, \infty) \to \mathbb{R}$, $f(x) = \ln(1+x)$ şi să se precizeze mulţ. pe care este valabilă dezv. găsită.

(b) $\sum_{n\geq 1} \frac{(-1)^{n+1}}{n} = ?$

SOL:

(a) $f'(x) = \frac{1}{1+x}$, f indefinit deriv pe $(-1, \infty)$.

$$\frac{1}{1+x} = \frac{1}{1-(-x)} = \sum_{n\geq 0} (-x)^n, \forall -x \in (-1,1) \Leftrightarrow x \in (-1,1)$$

$$f'(x) = 1 - x + x^2 - x^3 + x^4 - \dots, \forall x \in (-1,1)$$

$$\Rightarrow \underset{+C}{f(x)} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots =$$

$$= \sum_{n\geq 1} \frac{(-1)^{n+1} \cdot x^n}{(n+1)!}$$

$$x = 0 \Rightarrow \underset{0}{f(0)} + C = 0 \Rightarrow C = 0 \Rightarrow$$

$$\Rightarrow f(x) = \sum_{n\geq 0} \frac{(-1)^n \cdot x^{n+1}}{(n+1)!}, \forall x \in (-1,1) \quad (2)$$

b) $\sum_{n\geq 1} \frac{(-1)^{n+1}}{n} = ?$

$$\text{Din } (2) \Rightarrow x = 1 \Rightarrow \sum_{n\geq 0} \frac{(-1)^n}{n+1} = f(1) = \ln 2$$

$$= \sum_{n\geq 0} (-1)$$

5. • $f: \mathbb{R} \to \mathbb{R}$, $f(x) = \operatorname{arctg} x$

• $f: [-1,1] \to \mathbb{R}$, $f(x) = \arcsin x$.

a) $f'(x) = \frac{1}{1+x^2}$, f indefinit deriv.

$$\frac{1}{1+x^2} = (1+x^2)^{-1} \underset{\substack{x \leftarrow x^2 \\ \alpha = -1}}{=} 1 + \sum_{n\geq 1} \frac{(-1)\cdot(-2)\cdot}{}$$

$$\cdot \frac{(-n)}{n!} \cdot (x^2)^n =$$

$$= 1 + \sum_{n\geq 1} (-1)^n \cdot \frac{\cancel{n!}}{\cancel{n!}} \cdot x^{2n} =$$

$$= 1 + \sum_{n\geq 1} (-1)^n \cdot x^{2n}, \forall x^2 \in (-1,1)$$

$$= 1 - x^2 + x^4 - x^6 + \dots, \forall x \in (-$$

$$\Rightarrow f(x) + C = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

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$\Rightarrow f(x) + C = x + \sum_{n\geq 1} (-1)^n \cdot \frac{x^{2n+1}}{2n+1}$.

$x = 0 \Rightarrow \underbrace{f(0)}_{0} + C = 0 \Rightarrow C = 0$

$\Rightarrow f(x) = x + \sum_{n\geq 1} \frac{(-1)^n \cdot x^{2n+1}}{2n+1}, \forall n$

$= \sum_{n\geq 0} \frac{(-1)^n \cdot x^{2n+1}}{2n+1}$.

- $f: [-1, 1] \to \mathbb{R}$

$f(x) = \arcsin x$.

$f'(x) = \frac{1}{\sqrt{1-x^2}}$, f indef. deriv. pe $(-1,1)$

$(1-x^2)^{-\frac{1}{2}} = (1 + (-x^2))^{-\frac{1}{2}} =$

$= 1 + \sum_{n\geq 1} \frac{(-\frac{1}{2}) \cdot (-\frac{3}{2}) \cdots (-\frac{2n-1}{2})}{n!} \cdot (-x^2)^n$

$\forall x^2 \in (-1,1)$

$x \in (-1,1)$

$= 1 + \sum_{n\geq 1} \frac{\cancel{(-1)^n} \cdot (2n-1)!!}{2^n \cdot n!} \cdot \cancel{(-1)^n} \cdot x^{2n} =$

$= 1 + \sum_{n\geq 1} \frac{(2n-1)!!}{(2n)!!} \cdot x^{2n}$

$\overset{\int}{\Rightarrow} f(x) + C = x + \sum_{n\geq 1} \frac{(2n-1)!!}{(2n)!!} \cdot \frac{x^{2n+1}}{2n+1}$

$x = 0 \Rightarrow \underbrace{f(0)}_{0} + C = 0 \Rightarrow C = 0 \Rightarrow$

$\Rightarrow f(x) = x + \sum_{n\geq 1} \frac{(2n-1)!! \cdot x^{2n+1}}{(2n)!! \cdot (2n+1)}$

$\forall x \in (-1,1)$

Analiza topologică a unei mulțimi pe $\mathbb{R}^n$.

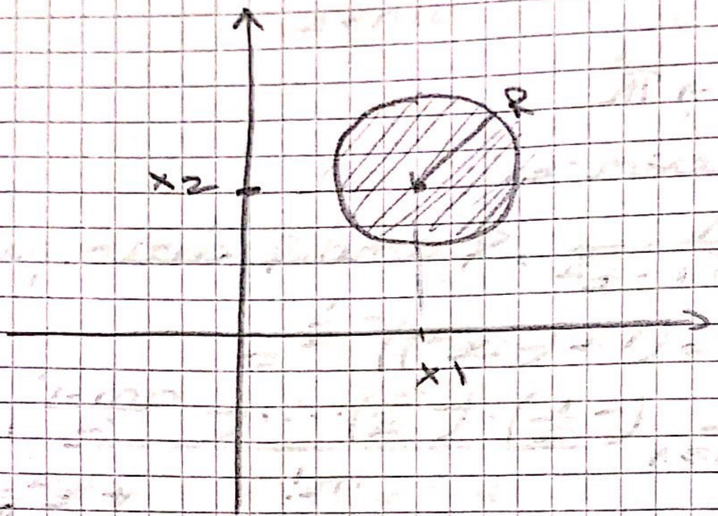
$\|\cdot\| \to$ norma euclidiană

Def. $x \in \mathbb{R}^n$, $r > 0$, $B(x,r) = \{y \in \mathbb{R}^n \mid \|y-x\| < r\} \to$ bila deschisă de centru x și raza r.

$(\mathbb{R})$: $B(x,r) = (x-r, x+r)$

$(\mathbb{R}^2)$: $\underset{(x_1,x_2)}{B(x,r)} = \{ \underset{(y_1,y_2)}{y} \in \mathbb{R}^2 \mid \|(y_1,y_2)-(x_1,x_2)\| < r \}$

$= \{ y \in \mathbb{R}^2 \mid (y_1-x_1)^2 + (y_2-x_2)^2 < r^2 \}$



Ex: ① $A = \{ (x,y) \in \mathbb{R}^2 \mid x^2+y^2 < 1, y \geq 0 \} \cup \{ (0, -\frac{n}{n+1}) \mid n \in \mathbb{N} \}$

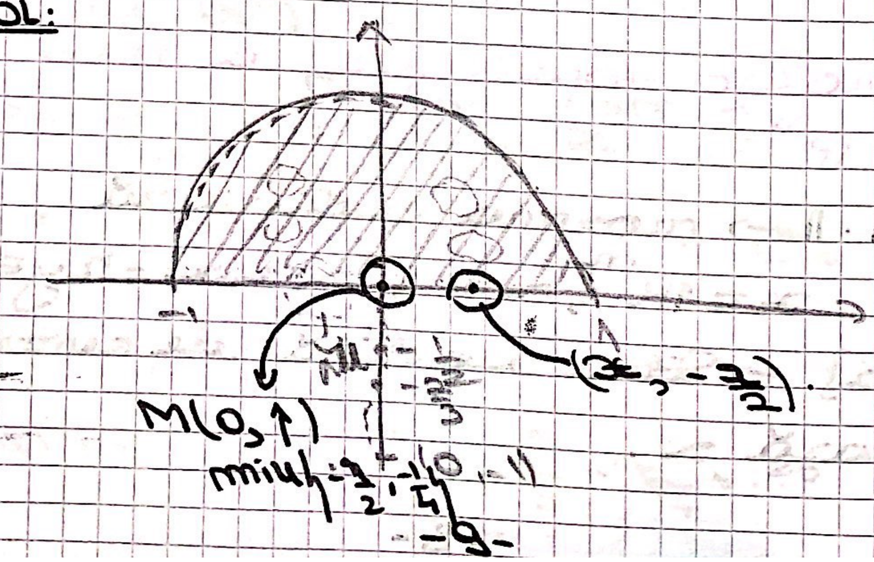
② $B = \{ (x,y) \in \mathbb{R}^2 \mid 0 \underset{\leq}{<} x \underset{\leq}{<} 1, 0 \underset{\leq}{<} y \underset{\leq}{<} 1 \} \setminus \{ (\frac{1}{n}, \frac{1}{n}), n \in \mathbb{N} \}$

$\mathring{A}$, A', $\bar{A}$, $Fr\,A$, A m.d, A m.f, A m.born

③ $C = \{ (x,y) \in \mathbb{R}^2 \mid 0 < x^2+y^2 \leq 1 \}$

SOL:

①



$\mathring{A} \subseteq A$

$\mathring{A} = \{(x,y) \in \mathbb{R}^2 \mid \exists r>0$ cu $B((x,y),r) \subseteq A\}$

$= \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2<1, y>0\}$

$\hookrightarrow \forall (x,y) \in \mathbb{R}^2$ cu $x^2+y^2<1, y>0$, evid

$\exists r>0$ a.î. $B((x,y),r) \subseteq A$.

$\hookrightarrow \forall (x,y) \in A$ de forma

$(x,0), x \in [-1,1]$

$(0, -\frac{n}{n+1}), n \in \mathbb{N}$. | $\forall r>0$, $B((x,y),r) \not\subseteq A$.

- $A' = \{(x,y) \in \mathbb{R}^2 \mid \forall V \in \mathcal{V}_{(x,y)}, V \cap A = \infty$ de el$\}$
- P $\forall (x_n)_n \subseteq A$ cu $x_n \xrightarrow{n} x \Rightarrow x \in A'$

$\bar{A} = A \cup A'$

$A' = \{(x,y) \mid x^2+y^2 \le 1, y \ge 0\} \cup \{(0,-1)\}$

$\hookrightarrow \forall (x,y) \in \mathbb{R}^2$ cu $x^2+y^2 \le 1, y \ge 0$

$\forall V \in \mathcal{V}_{(x,y)}$ avem că

$V \cap A = \infty$ de el.

$\bar{A} = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \le 1, y \ge 0\}$

$\cup \{(0, -\frac{n}{n+1}) \mid n \in \mathbb{N}\}$.

A deschisă NU ($A \neq \mathring{A}$)

A închisă NU ($A \neq \bar{A}$)

A mărg DA $A \subseteq B((0,0),2)$

A compactă NU

A conexă NU

D1 =

D2 =

A neconexă dacă

$\exists$ D1, D2 deschise

$(D1 \cap A) \cap (D2 \cap A) = \emptyset$

$D1 \cap A \neq \emptyset$ $(D1 \cap A) \cup (D2 \cap A) = A$

$D2 \cap A \neq \emptyset$

$D_1 = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 < 2,\ y > -\frac{1}{8}\}$

$D_2 = B((0,-1), \frac{1}{4})$.

$\Rightarrow$ nu e conexă.

$\text{Fr}\,A = \bar{A} \setminus \mathring{A}$